

Combining Technical Indicators and Genetic Algorithms for Short-Term Machine Learning Prediction of Mini-Index Futures

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Abstract. This article investigates short-term trend forecasting in the mini-index futures market using a hybrid framework that integrates technical analysis, feature selection, and evolutionary hyperparameter optimization. Five-minute intraday data obtained from the BovDBV2 database are used to construct a set of technical indicators that capture short-term price dynamics. To address the redundancy and noise commonly found in high-dimensional financial data, a feature selection method is applied to reduce the original feature space to a compact and informative subset. Genetic algorithms are then employed to optimize the hyperparameters of three supervised learning models: Random Forest, Support Vector Machine, and Multilayer Perceptron. Experimental results indicate that genetic algorithm optimization consistently improves model performance when compared to a baseline Random Forest model without hyperparameter tuning, which achieved an accuracy of approximately 61.63%. Within the proposed framework, the optimized Support Vector Machine and Multilayer Perceptron models reach test accuracies of 65.61% and 65.85%, respectively, while maintaining balanced $F1 - scores$ across up-trend and downtrend classes. These results demonstrate the effectiveness of combining feature selection with evolutionary hyperparameter optimization for short-term financial time series forecasting in derivatives markets.

Keywords: Neural Networks · Machine Learning · Genetic Algorithm · Intraday Trading · Financial Forecasting.

1 Introduction

Futures contracts are financial instruments with high trading volume in the intraday market, due to their combination of high liquidity and direct exposure to price fluctuations. They define agreements for the purchase or sale of an underlying asset at a predetermined price, with settlement on a future date [13].

In Brazil, the futures contract such as Mini Index (WIN), listed on the B3 (Brazilian Stock Exchange)⁴, concentrates a significant portion of the daily intraday volume, drawing attention as a particularly relevant instrument for short-term forecasting research. These instruments are heavily traded in day trading; however, identifying reliable patterns for forecasting over such short periods has been a challenge due to the inherent volatility and dynamic behavior of the time series. This complexity motivates researchers and professionals to develop more robust and adaptable modeling approaches.

One of the early approaches to predicting financial time series was the use of static models like ARIMA [22]; however, because these models assume simple linearity and temporal dependence, they do not fit well with the highly complex and dynamic structure of the current financial market. More recently, with the growing intersection between computing and finance and the increasing availability of data, methods such as Machine Learning (ML) [14] and Recurrent Neural Networks (RNN) [10] have emerged, which impose fewer distributional constraints and can capture non-linear relationships directly from price series.

The best-known ways to "feed" the input to these models is by using mathematical values, such as historical price information, market data, and technical analysis indicators like moving averages, volatility measures, and others. By using this data with advanced models, we can try to predict market fluctuations [17]. However, to increase the accuracy of the prediction, one of the most critical steps in building more robust machine learning models is feature selection, which involves identifying and retaining only the most relevant attributes. Since not all attributes contribute equally to predictive performance, techniques such as filtering, encapsulation, or embedding are essential to minimize overfitting and improve model generalization.

Technical indicators derived from price series — moving averages, volatility measures, among others — serve as input variables in data-driven forecasting models [17]. Including a large number of them, however, tends to introduce redundancy, which can destabilize model training. Feature selection addresses this by restricting the input space to the most informative variables. That said, a reduced feature set does not compensate for poorly chosen hyperparameters. Genetic Algorithms (GA) [6] have been used in this context to search for configurations that improve generalization beyond what manual tuning typically achieves.

In parallel, hyperparameter optimization plays a critical role in improving model performance, particularly in flexible models such as Random Forest (RF) and Multilayer Perceptron (MLP). Evolutionary approaches, such as Genetic Al-

⁴ <https://www.b3.com.br/>

gorithms (GA), are well suited for this task due to their ability to explore complex and non-convex search spaces [9] and avoid local optima, unlike gradient-based methods [6].

The main objective of this article is to investigate how the combined application of feature selection and evolutionary optimization strategies adapts to more expressive and sensitive learning models for intraday financial forecasting. By comparing models that applied these techniques with a Random Forest (RF) baseline model, we aim to demonstrate that these methods aid in generalization to classifiers with greater representational power and exhibit a lower risk of overfitting. Here, we expand on previous research by jointly optimizing feature subsets and hyperparameters for Support Vector Machines (SVMs) and Multilayer Perceptrons (MLPs) using a unified genetic framework, systematically evaluating robustness and generalization in forecasting financial time series.

The remainder of this article is organized as follows: Section 2 presents the theoretical background. Section 3 reviews related work. Section 4 describes the methodology. Section 5 presents and discusses the results. Section 6 concludes and outlines future directions.

2 Theoretical Concepts

In this section, we aim to contextualize our research by presenting the technical analysis indicators that are used as input to the models, followed by an exploration of the feature selection techniques employed.

2.1 Technical Analysis Indicators

Technical analysis indicators are quantitative tools derived from historical asset data that are used to predict future market movements. These indicators help traders identify trends, momentum, and potential reversal points, offering a systematic approach to interpreting market fluctuations.

Moving Average The moving average is a fundamental statistical technique used to smooth the volatility inherent in time series data [12]. Different forms of Moving Averages, such as the Simple Moving Average (SMA) and the Exponential Moving Average (EMA), provide varying degrees of sensitivity to recent price changes [4]. The use of moving averages in technical analysis helps traders to filter out the 'noise' in market data, making it easier to identify support and resistance levels and to confirm trend directions.

Standard Deviation Standard deviation is a statistical metric that quantifies price volatility by measuring how much an asset's price deviates from its mean over a specific period [1]. In technical analysis, it is calculated by first determining the moving average (mean) of closing prices and then calculating the squared differences between each price and that average. The average of these squared differences produces the variance, and the square root of the variance produces the standard deviation.

2.2 Machine Learning for Price Prediction

Three supervised learning algorithms are evaluated. Random Forest constructs multiple decision trees and aggregates their outputs, which provides some robustness to noise and allows nonlinear relationships to be captured [3]. Support Vector Machines identify a separating boundary in the feature space, extended to nonlinear cases through kernel functions. Multilayer Perceptrons are feed-forward neural networks capable of approximating complex functions through compositions of linear transformations and nonlinear activations [2, 8]. Since model performance is sensitive to hyperparameter configuration, evolutionary optimization is applied to all three classifiers to improve generalization on unseen data.

3 Related Work

Machine learning methods have been widely applied to financial forecasting, often showing better results than traditional econometric models [10]. Ballings et al. [2], in a study involving data from thousands of European companies, evaluated several classifiers and found that Random Forests and SVMs performed better than logistic regression across most of the tested scenarios [4]. These results pointed to the relevance of nonlinear methods for capturing patterns in financial return data, where linear assumptions tend to hold poorly.

M. Ballings et al. [2] evaluated several classifiers using data from European companies and found that Random Forests and SVMs performed better than logistic regression across most tested scenarios. The study focused on longer-term return prediction rather than intraday forecasting. In the present work, the same classifiers are evaluated, but applied to 5-minute WIN futures data, where the noise level and pattern persistence differ considerably from the setting studied by Ballings et al.

F. Ecer [8] investigated how network structure and training configuration affect MLP performance on financial time series, finding that task-specific adaptations produced better out-of-sample results than default settings. The study did not combine architectural search with feature selection. Here, both aspects are addressed jointly through a genetic algorithm applied to the same optimization process.

Naik and Mohan [18] applied the Boruta algorithm to filter technical indicators before training an ANN on NSE data, reducing prediction error by 12% compared to the full feature set. Their work treated feature selection as an isolated step, independent of model configuration. The present work extends this idea by combining feature selection with hyperparameter optimization in a unified pipeline.

Ding et al. [7] combined back-propagation with a genetic algorithm and reported better generalization compared to standard gradient-based training on complex datasets. The study focused on the optimization of neural network weights rather than hyperparameters. In contrast, this work uses GA to search

over the hyperparameter space of three distinct classifiers applied to intraday financial data.

Taken together, these studies address feature selection and hyperparameter optimization as separate problems. This work evaluates their combined effect on intraday trend prediction using WIN futures data from the Brazilian market, comparing Random Forest, SVM and MLP under the same experimental conditions.

4 Methodology

This section presents the methodological steps adopted in the study. Figure 1 summarizes the experimental flow, including data acquisition, feature construction, feature selection, and model evaluation.

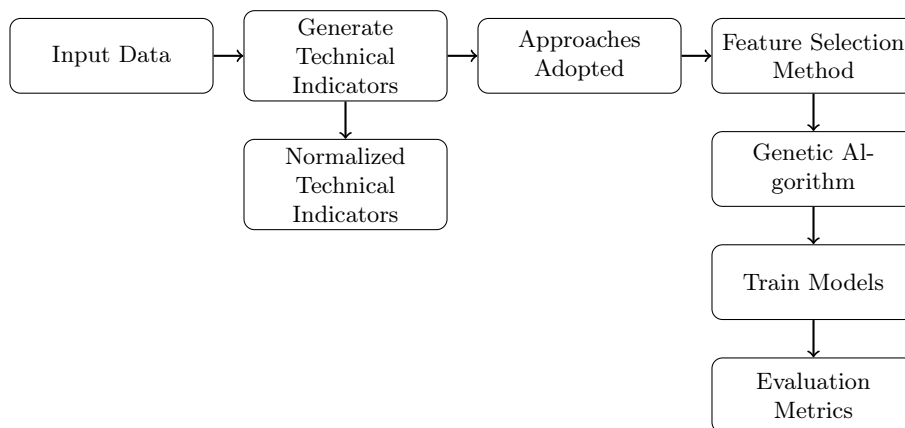


Fig. 1. Overview of the proposed methodology.

4.1 Input Data

The data used in this study were obtained from the BovDbV2 database [20], an updated version of BovDB [5]. For this work, we used the 5-minute intraday records available from January to June 2024. This time interval was adopted because it allows the analysis of short-term price variation while avoiding the higher noise level usually found in tick-by-tick data.

The study focused on the Mini Index futures contract (WIN), since it is one of the most actively traded derivatives on B3 and is frequently used in short-term trading. To keep the analysis aligned with the most liquid contracts in each period, three rollover intervals were considered: WING24 to WINJ24 (January–February), WINJ24 to WINM24 (March–April), and WINM24 to WINQ24 (May–

June). After preprocessing, this selection resulted in approximately 120 trading days and more than 15,000 instances.

The target variable was defined as a binary label indicating the direction of the next relevant price movement. To obtain this label, we applied the peak-and-trough detection procedure described in [21] to identify local extrema in the series. Each instance received label 1 when the following movement was classified as an uptrend and label 0 when it was classified as a downtrend. This definition allowed the task to be treated as a binary classification problem.

4.2 Generation of Technical Analysis Indicators

Technical indicators were computed directly from the 5-minute price series. Since the objective was to capture short intraday movements, only short windows were considered. The adopted periods were 3, 5, 7, and 9 intervals. Based on these windows, three types of indicators were generated: Simple Moving Average (SMA), Exponential Moving Average (EMA), and Standard Deviation (STD).

To capture changes in trend dynamics, we calculated the differences between the moving averages calculated over different time periods. For example, the difference between the 7-period SMA and the 3-period SMA is represented in Eq. (1):

$$\text{SMA}_{7-3} = \text{SMA}_7 - \text{SMA}_3, \quad (1)$$

Normalization scales features to a common range (typically $[0, 1]$) to improve model adaptability [19]. For every 5-minute interval, we applied a min-max scaling, formulated in Eq. (2):

$$\text{Feature}_{\text{norm}} = \frac{\text{Feature} - \text{Low}}{\text{High} - \text{Low}}. \quad (2)$$

Here, *High* and *Low* define the price limits within each 5-minute window. After these normalization steps, and combining them with standard daily pricing features—open, high, low, average, close, buy price, sell price, trading volume, and number of shares—the resulting dataset totals 66 features.

4.3 Feature Selection Method

This section will discuss how the feature selection step was performed. Since the complete dataset contained 66 features, a feature selection step was introduced to reduce redundancy and simplify the search space explored during model fitting.

Among the methods evaluated, Information Gain produced the most stable results across all training partitions and was therefore adopted for model training. This criterion estimates how much each feature reduces the uncertainty about the target class by identifying a compact and informative subset [16], selecting 7 features from the original 66.

The retained subset includes the 5–3 and 7–3 EMA differences, their normalized counterparts, the normalized 9-period SMA, the 9–3 EMA difference, and the normalized 7-period EMA.

This reduced subset was then adopted as the fixed input space for the Genetic Algorithm optimization stage. With fewer input variables, the training process becomes less affected by irrelevant information, and the search becomes easier to manage [15]. In practice, this also makes the fitness evaluation more stable across generations.

4.4 Genetic Algorithm Parameters

Genetic Algorithms (GAs) are evolutionary optimization methods based on selection, crossover, and mutation [9, 11]. In this study, they were used because hyperparameter tuning involves a discrete and nonlinear search space. By evaluating different candidate solutions over successive generations, GA can gradually retain the best configurations found during the search.

The GA optimization cycle comprises population initialization, fitness evaluation, parent selection, crossover, mutation, and elitist preservation of the best solution. To ensure methodological consistency and allow a direct comparison across experiments, the same GA configuration was adopted for all models. Parent selection was performed using tournament selection with size 3, while crossover and mutation were applied with fixed probabilities throughout the optimization process. The adopted configuration is summarized in Table 1.

Table 1. Genetic Algorithm parameters adopted in this study.

Parameter	Value
Number of generations (N_{gen})	1000
Population size (N_{pop})	10
Crossover probability (p_c)	0.80
Mutation probability (p_m)	0.50
Per-gene mutation probability	0.05
Tournament size	3
Elitism (Hall-of-Fame size)	1
Random seed	42

The fitness of each individual was computed exclusively on the training partition using stratified cross-validation, whereas the test partition was reserved for the final out-of-sample evaluation. For the Random Forest and Multilayer Perceptron models, fitness was defined as a weighted combination of training and validation accuracy in order to promote generalization while discouraging severe underfitting. The fitness function is given by:

$$\text{Fitness} = 0.4 \cdot \text{Acc}_{train} + 0.6 \cdot \text{Acc}_{val}, \quad (3)$$

The final fitness corresponds to the average value across all folds. After the optimization stage, the best hyperparameter configuration found by the GA was used to retrain each model on the full training set before the final evaluation on the test set.

5 Results

This section presents the results obtained in the experiments. For clarity, it is divided into two parts. The first describes the experimental setup and data partitioning procedure, and the second compares the predictive performance of the evaluated models.

5.1 Experimental Setup and Data Partitioning

To ensure a fair comparison, all models were evaluated under the same experimental conditions. Each algorithm used the same 7-feature input space selected by the Information Gain criterion Section 4.3, and the same Genetic Algorithm configuration described in Section 4.4.

Since the dataset presented class imbalance, a balanced sampling procedure was adopted for the training stage. We selected 1,500 uptrend instances and 1,500 downtrend instances, resulting in a training set with 3,000 observations. The remaining 12,057 instances were kept as an independent test set. A fixed random seed was used to make the sampling reproducible.

During the GA optimization stage, model fitness was evaluated using 5-fold cross-validation on the training set. After the best hyperparameter configuration was found, each model was retrained on the full 3,000-instance training set and then evaluated on the 12,057 test instances.

All experiments were executed on the same hardware platform, with a 12-core CPU and 16 GB of RAM, to allow runtime comparison. Under these conditions, the complete optimization and training procedure took about 2 hours for Random Forest, 3 hours for SVM, and nearly 18 hours for MLP.

5.2 Performance of the Models

Figure 2 shows the evolution of the Random Forest fitness during the GA optimization. The cross-validation accuracy increases in the first generations and then stabilizes, indicating convergence to a more stable region of the search space and Table 2 presents the search space and the final configuration selected for Random Forest.

Table 2. Search space optimized by GA for RF

Hyperparameter	Search Space	Optimal Value
<i>n_estimators</i>	[50, 500]	200
<i>max_depth</i>	[5, 50]	15
<i>min_samples_split</i>	[2, 20]	4
<i>min_samples_leaf</i>	[1, 20]	1

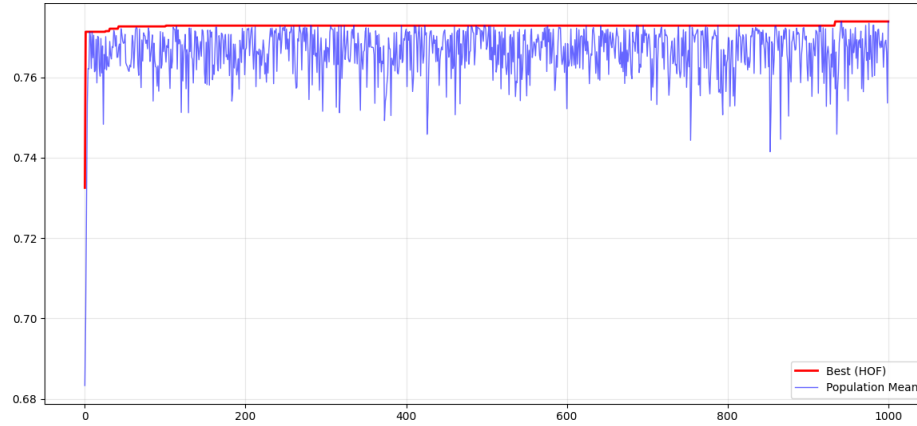


Fig. 2. Evolution of the fitness function during the Genetic Algorithm optimization of the Random Forest model, where the x-axis denotes the generation number and the y-axis represents the average cross-validation accuracy.

The optimized values for $n_estimators$, max_depth , $min_samples_split$, and $min_samples_leaf$ indicate that the GA favored a model with controlled complexity. The best configuration in the Hall of Fame (HOF) achieved an average cross-validation accuracy of 0.7739. And in the test set, a final accuracy of 0.6331. Performance by class remained relatively balanced, with F1 scores of 0.6169 for downtrends and 0.6479 for uptrends.

For the Support Vector Machine (SVM), we defined the kernel as a Radial Basis Function (RBF), as it is faster to train compared to functions like Polynomial (POLY) and captures trends more quickly.

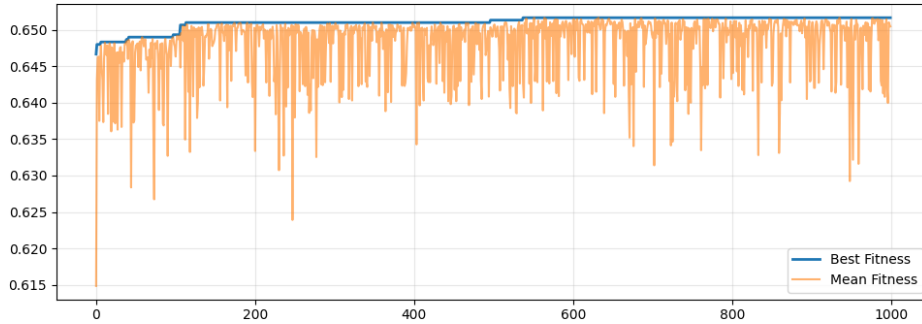
Figure 3 shows the evolution of the SVM model's performance throughout the hyperparameter optimization process using Genetic Algorithm (GA). Compared to Random Forest (RF), the SVM showed less tendency to overfit the training data, which may be related to its greater generalization capacity in high-dimensional spaces. The best fitness value obtained was 0.6517, indicating stable behavior during the optimization process.

Table 3 presents the hyperparameters C and γ and the values obtained from the search space optimization by GA. While C balances maximizing the margin and minimizing the error, γ determines the local radius of influence. The combination of these parameters affects the flexibility and generalization of the SVM model.

For the test set of 12,057 unseen instances, the model achieved an accuracy of 0.6561, outperforming the Random Forest. Like the Random Forest, the SVM achieved balanced performance across trend classes, with greater effectiveness in detecting uptrends, resulting in F1 scores of 0.6308 and 0.6781 for downtrends and uptrends, respectively.

Table 3. Search space optimized by the GA for SVM

Hyperparameter	Search Space	Optimal Value
C	$\log_{10}(C) \in [-3, 3]$	9.75×10^2
γ	$\log_{10}(\gamma) \in [-5, 1]$	1.28×10^{-5}
Kernel	Fixed	RBF

**Fig. 3.** Evolution of the fitness function during the Genetic Algorithm optimization of the SVM model (RBF kernel), where the x-axis denotes the generation number and the y-axis represents the average cross-validation accuracy.

The Multilayer Perceptron classifier was optimized using GA, following our methodology. This model is a type of artificial neural network composed of several layers of interconnected neurons. It is widely used in machine learning tasks, such as classification, regression, and pattern recognition, due to its ability to learn non-linear relationships in data, making it particularly suitable for financial time series classification problems.

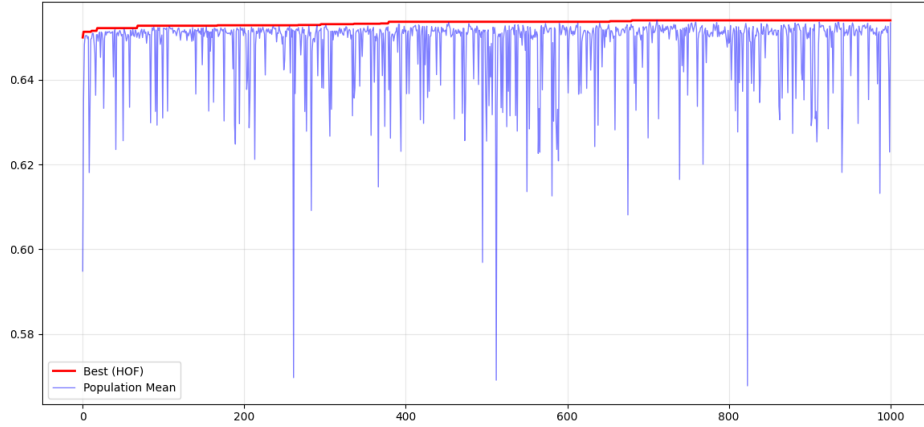
For the MLP, the GA optimized both architectural and training hyperparameters, including the number of hidden neurons, learning rate, L_2 regularization, dropout rate, batch size, activation function, and optimizer.

Figure 4 presents the evolution of MLP fitness over 1,000 generations. The best individual's fitness increases gradually and then stabilizes. The average fitness shows greater variation across generations, which is expected during the search process.

The GA optimization of the MLP produced fitness values in a relatively narrow interval, ranging from 0.6499 to 0.6540 over 1,000 generations. This limited variation suggests that the search process converged to a stable region of the hyperparameter space, without large oscillations in performance. The best configuration reached a cross-validation accuracy of 0.6540. Table 4 summarizes the optimized hyperparameters, their search ranges, and the final values selected by the GA.

Table 4. Search space optimized by the GA for MLP

Hyperparameter	Search Space	Optimal Value
Hidden neurons	[5, 500]	44
Learning rate	$[10^{-8}, 10^{-2}]$	4.70×10^{-3}
L_2 regularization	$[10^{-8}, 10^{-2}]$	4.48×10^{-6}
Dropout rate	[0.00, 0.10]	0.07
Batch size	$\{16, \dots, 128\}$	80
Activation function	Fixed	Tanh
Optimizer	Fixed	RMSprop

**Fig. 4.** Evolution of the fitness function during the Genetic Algorithm optimization of the MLP model, where the x-axis denotes the generation number and the y-axis represents the average cross-validation accuracy.

The selected configuration kept the hidden layer at a moderate size, which is consistent with the need to control model complexity when dealing with noisy intraday financial data. The optimization also selected a small L_2 regularization term and a low learning rate, indicating that a more conservative training process was beneficial for this problem.

With RMSprop and Hyperbolic Tangent activation, the MLP achieved a test accuracy of 0.6585. This was the highest result among the evaluated models, slightly exceeding the performance of the SVM and remaining clearly above that of Random Forest. In the class-specific analysis, the model showed better performance for uptrends, with an F1-score of 0.6753, compared with 0.6397 for downtrends.

To evaluate the sensitivity of the Multilayer Perceptron to the optimization strategy, a second model test experiment was conducted using the same search space configuration shown in the previous test, as per Table 4, with the only

modification being the optimization algorithm employed during training, which was Adam.

The best individual found by the GA in this configuration specified a hidden layer with 89 neurons, a learning rate of 1.94×10^{-3} , an L_2 regularization coefficient of approximately 8.71×10^{-8} , a dropout rate of 0.01 and a batch size of 112, while keeping the hyperbolic tangent activation function as the nonlinear transformation in the hidden layer.

Table 5. Comparison of MLP performance using different optimizers optimized by the Genetic Algorithm

Metric	MLP + RMSprop	MLP + Adam
GA Fitness (min)	0.6499	0.6471
GA Fitness (max)	0.6540	0.6514
Test Accuracy	0.6585	0.6544
Precision	0.6755	0.6783
Recall	0.6789	0.6623
F1-score	0.6772	0.6702
AUC-ROC	0.7032	0.7023
AUC-PR	0.7029	0.7022

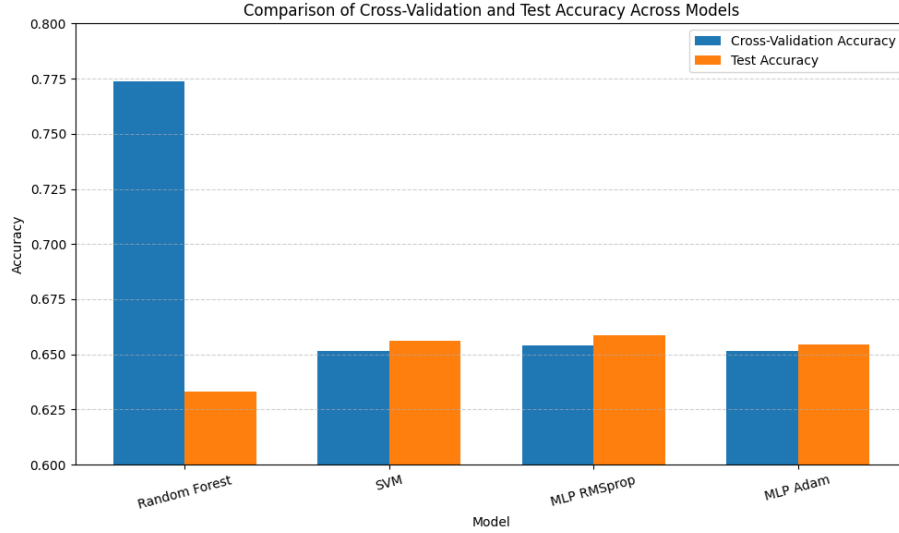


Fig. 5. Comparison between cross-validation and test accuracies obtained for the evaluated models.

Table 5 compares two MLP models optimized with RMSprop and Adam, focusing on min/max GA fitness, final test accuracy, and metrics like AUC-ROC and AUC-PR.

Table 5 shows both optimizers have similar probabilistic performance, with nearly identical AUC-ROC and AUC-PR values. The RMSprop-based MLP has slightly higher cross-validation fitness and test accuracy, while Adam yields a marginally higher F1-score, indicating the optimizer mainly influences the precision-recall trade-off rather than causing significant performance differences in the GA search space.

Figure 5 puts the performance of all three algorithms into perspective. Random Forest showed the highest accuracy in cross-validation, but the difference between validation and test results was greater. This is because tree-based algorithms often suffer from overfitting to the noise inherent in market microstructures, leading to exactly this type of generalization failure. SVM and MLP based on RMSprop showed closer results in these two stages; the difference in generalization between training and testing is not as large, indicating more consistent generalization on unseen data. In this comparison, MLP (RMSprop) and SVM obtained the best results in the tests.

6 Conclusion

Previous results in the Brazilian derivatives market [21] indicated that selecting intraday Mini-Index technical indicators can reduce computational cost and improve baseline generalization. However, feature selection alone was not sufficient to obtain the best predictive performance. Without hyperparameter optimization, the baseline Random Forest reached a test accuracy of 61.63%.

To improve these results, we introduced a Genetic Algorithm for hyperparameter optimization and evaluated classifiers with different model structures. The optimized Random Forest achieved the highest cross-validation accuracy, but its performance dropped more clearly on the test set. In contrast, SVM and MLP showed more consistent generalization on unseen data.

These results indicate that, in this problem, the SVM and MLP models handled noisy and nonlinear intraday patterns more effectively than Random Forest. Among the evaluated models, the MLP with RMSprop achieved the highest test accuracy, reaching 65.85%.

Overall, the results suggest that combining feature selection with hyperparameter optimization is more effective than using a default configuration after feature reduction alone. Although this framework was applied here to short-term index forecasting, the same strategy may also be useful in other predictive tasks involving noisy and non-stationary data.

As a continuation of this study, future experiments may include additional technical indicators, such as RSI, MACD, and volatility- or volume-based variables, in order to verify whether the predictive performance can be improved with a broader representation of market behavior.

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